

Security Incident Report

Windows Endpoint Intrusion Simulation

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Executive Summary

This report documents a controlled multi-stage Windows endpoint intrusion simulation conducted in an isolated lab environment to demonstrate Security Operations Center (SOC) detection, correlation, and investigative workflow capabilities.

The simulated adversary achieved the following:

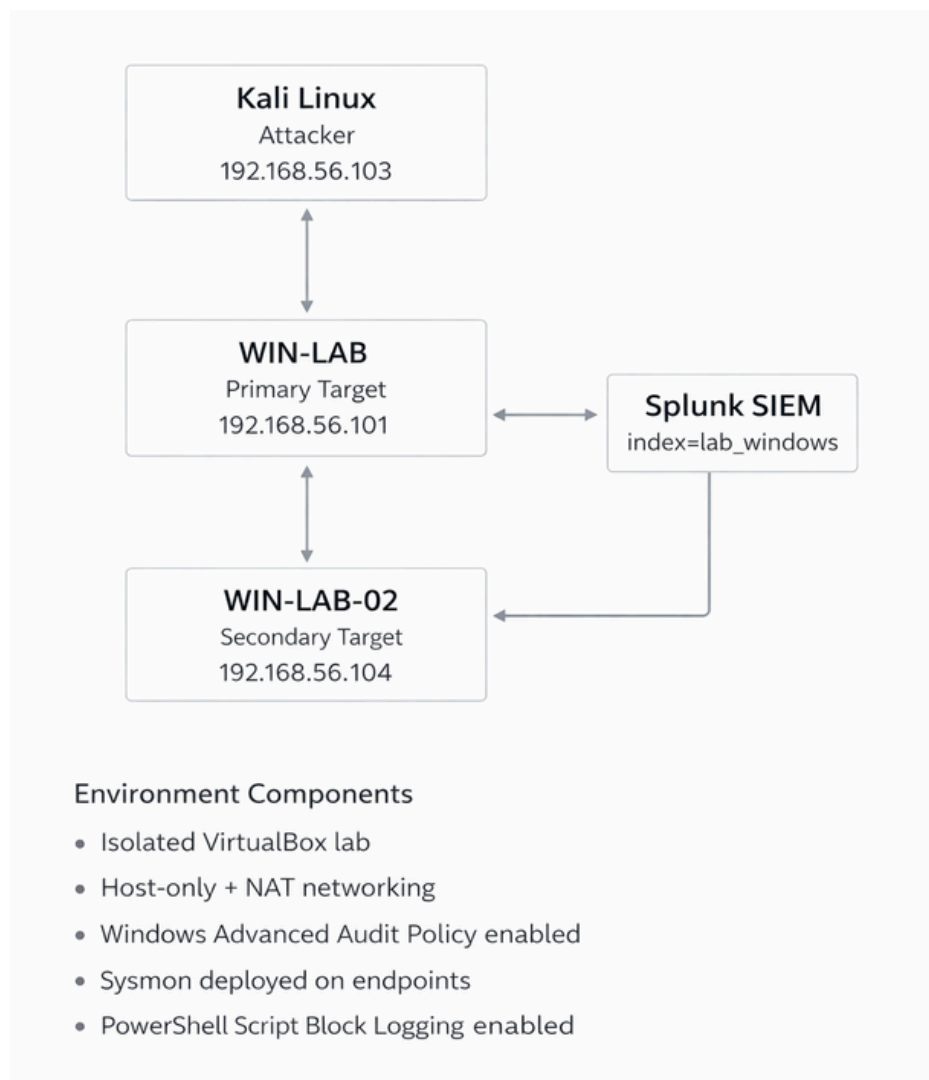
- Initial access via RDP brute-force authentication
- Administrative privilege assignment
- Persistence through local account creation
- Encoded PowerShell execution
- Security log tampering
- Lateral movement to a secondary host

All activity was reconstructed using Windows Security Logs (4625, 4624, 4672, 4720, 4732, 1102), PowerShell Script Block Logging (4104), Sysmon telemetry, and Splunk-based detection analytics.

This case study demonstrates Tier 1 / Tier 2 SOC investigative methodology and detection engineering fundamentals.

Modeled Risk Level: High
Detection Confidence: High

Lab Architecture



Incident Impact Assessment

Impact Area	Risk Level
Confidentiality	High
Integrity	High
Availability	Medium
Detection Evasion	High

The combination of credential compromise, administrative persistence, and security log tampering represents a high-impact intrusion path. In an enterprise environment, this level of access would require immediate containment and credential reset procedures.

Attack Timeline Overview

Phase	Activity	Key Event IDs
Initial Access	RDP brute-force authentication	4625, 4624
Privilege Escalation	Administrative token assignment	4672
Persistence	Local account creation & admin group addition	4720, 4732
Execution	Encoded PowerShell activity	4688, 4104
Defense Evasion	Security log cleared	1102
Lateral Movement	SMB / remote authentication to WIN-LAB-02	4624, 4672

Time Normalization Notice

Splunk stores and normalizes timestamps in UTC during log ingestion. Windows Event Viewer displays local system time. Minor date differences between Event Viewer and Splunk reflect timezone normalization during ingestion and do not affect event sequencing or analytical accuracy.

Phase 1 - Initial Access

Multiple failed RDP authentication attempts (Event ID 4625) were observed originating from 192.168.56.103 against the local administrative account.

Following repeated failures, a successful authentication event (4624) was recorded from the same source IP, confirming credential compromise via brute-force technique.

New Search

Save As>Create Table View>Close

Index=lab_windows EventCode=4625
| table _time Computer IPAddress TargetUserName LogonType

Time range: All time

11 events (before 2/19/26 7:00:11.000 PM)No Event Sampling

Job

EventsPatternsStatistics (11)Visualization

Show: 20 Per PageFormatPreview: On

_time	Computer	IPAddress	TargetUserName	LogonType
2026-02-19 17:01:22	WIN-LAB	127.0.0.1	vboxuser	2
2026-02-19 17:01:22	WIN-LAB	192.168.56.103	itsadmin	3
2026-02-19 17:01:22	WIN-LAB	192.168.56.103	itsadmin	3
2026-02-19 17:01:22	WIN-LAB	192.168.56.103	itsadmin	3
2026-02-19 17:01:22	WIN-LAB	192.168.56.103	itsadmin	3
2026-02-19 17:01:22	WIN-LAB	192.168.56.103	itsadmin	3
2026-02-19 17:01:22	WIN-LAB	192.168.56.103	itsadmin	3
2026-02-19 17:01:22	WIN-LAB	192.168.56.103	itsadmin	3
2026-02-19 17:01:22	WIN-LAB	192.168.56.103	itsadmin	3
2026-02-19 17:01:22	WIN-LAB	192.168.56.103	itsadmin	3
2026-02-19 17:01:22	WIN-LAB	192.168.56.103	itsadmin	3

Remote authentication brute-force over network (Logon Type 3)

splunk-enterpriseApps

AdministratorMessagesSettingsActivityHelpFind

SearchAnalyticsDatasetsReportsAlertsDashboardsModules

New Search

Save As>Create Table View>Close

Index=lab_windows EventCode=4625
| stats count by IPAddress
| sort - count

Time range: All time

11 events (before 2/19/26 7:03:03.000 PM)No Event Sampling

Job

EventsPatternsStatistics (2)Visualization

Show: 20 Per PageFormatPreview: On

IPAddress	count
192.168.56.103	10
127.0.0.1	1

Event ID 4625 – Failed network logon from 192.168.56.103 By Count

Filtered Log: Security; Source: ; Event ID: 4624. Number of events: 22

Keywords

Date and Time

Source

Event ID

Task Category

Q_Audit Success

2/18/2026 8:19:32 PM

Microsoft Windows security auditing.

4624

Logon

Event 4624, Microsoft Windows security auditing

General

Details

An account was successfully logged on.

Subject:

Security ID: NULL SID
Account Name: -
Account Domain: -
Logon ID: 0x0

Logon Information:

Logon Type: 3
Restricted Admin Mode: No
Virtual Account: No
Elevated Token: No

Impersonation Level: Impersonation

New Logon:

Security ID: WIN-LAB\itsadmin
Account Name: itsadmin
Account Domain: WIN-LAB
Logon ID: 0x67FE1C
Linked Logon ID: 0x0
Network Account Name: -
Network Account Domain: -
Logon GUID: {00000000-0000-0000-0000-000000000000}

Process Information:

Process ID: 0x0
Process Name: -

Network Information:

Workstation Name: kali
Source Network Address: 192.168.56.103

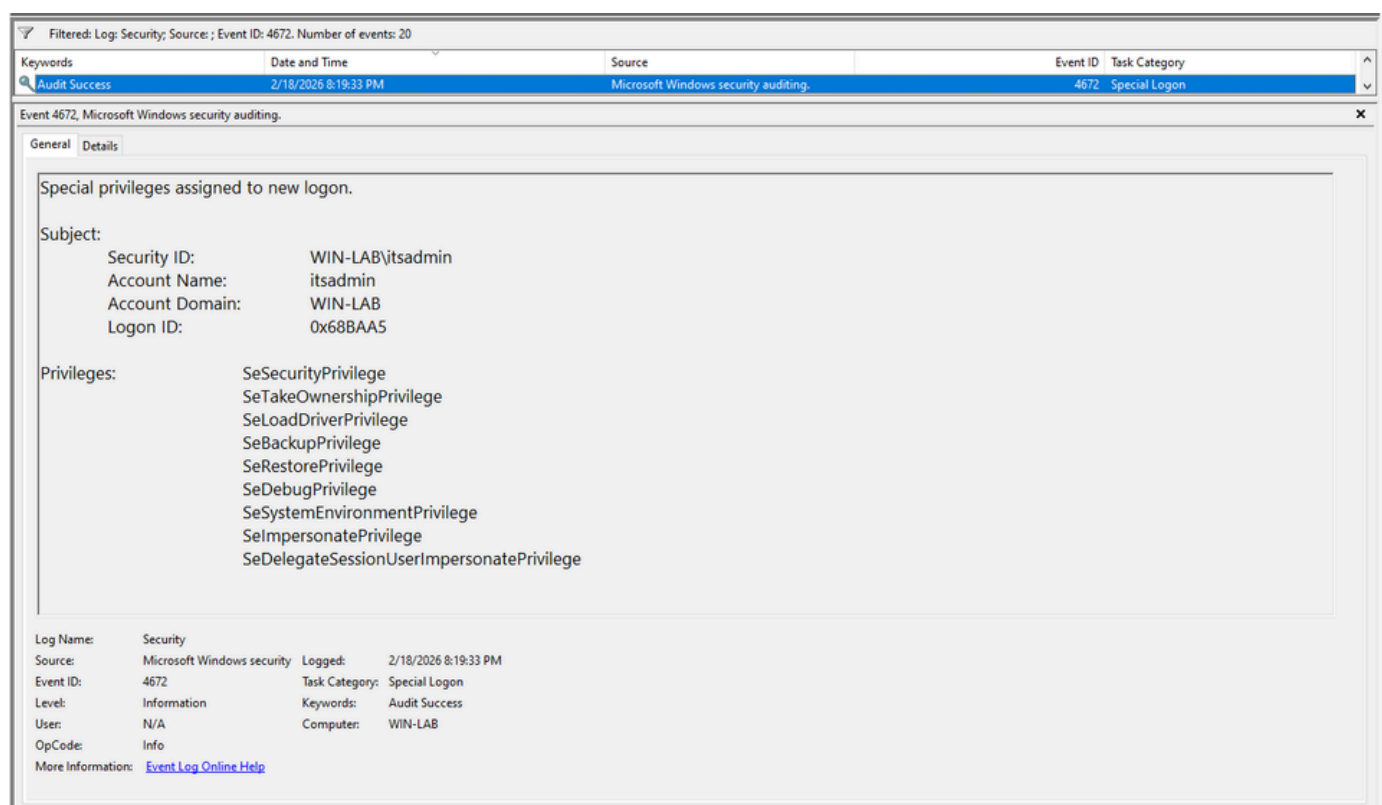
Analyst Assessment

Repeated authentication failures from 192.168.56.103 were observed over a sustained period and were followed approximately 10 minutes later by a successful network logon to the local administrative account. The sequence strongly indicates brute-force credential compromise.

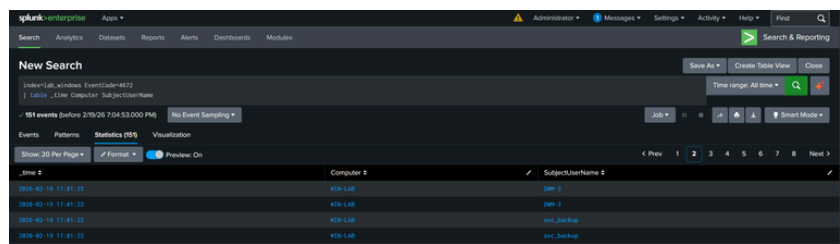
Phase 2 - Privilege Escalation

Following successful authentication, a privileged logon event (Event ID 4672) was recorded on the compromised host (WIN-LAB). This event indicates that administrative-level privileges were assigned to the authenticated session, confirming elevated access.

Event ID 4672 is generated when special privileges such as SeDebugPrivilege or SeBackupPrivilege are granted to a logon session.



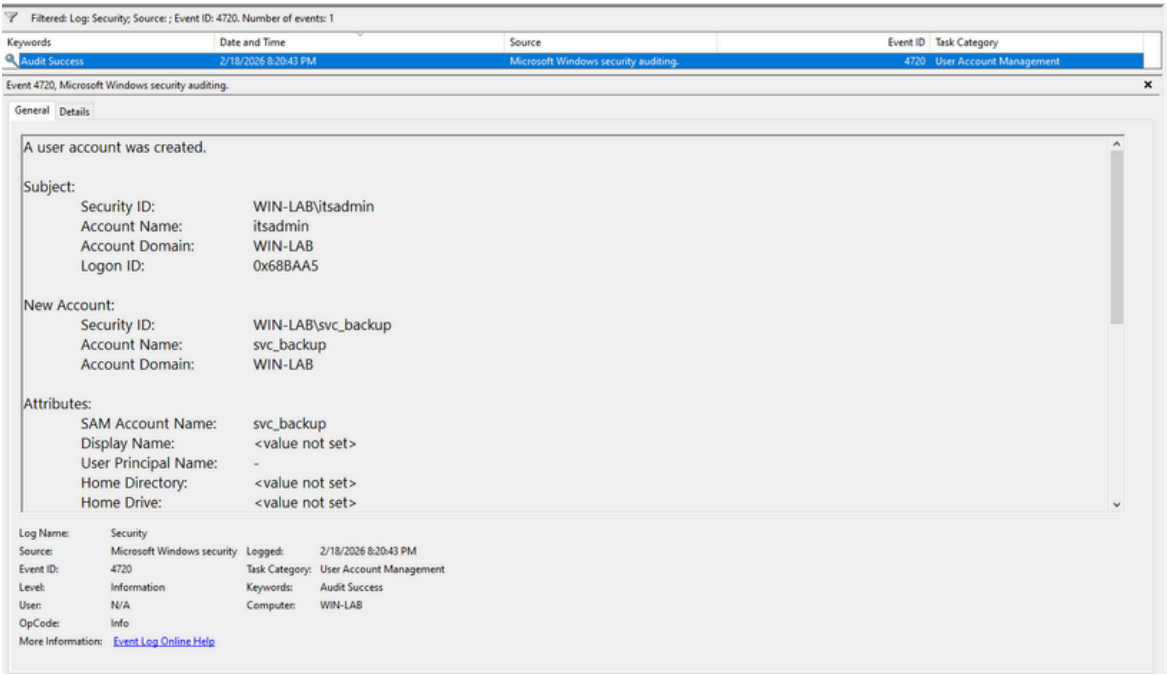
Windows Security Log – Event ID 4672
Special privileges assigned to new logon (administrative token granted).



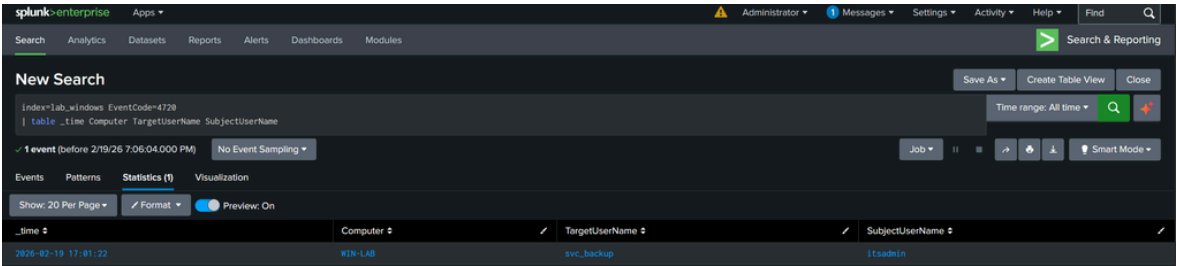
Splunk Detection – EventCode=4672
Privileged logon events observed on WIN-LAB following successful authentication.

Phase 3 – Persistence

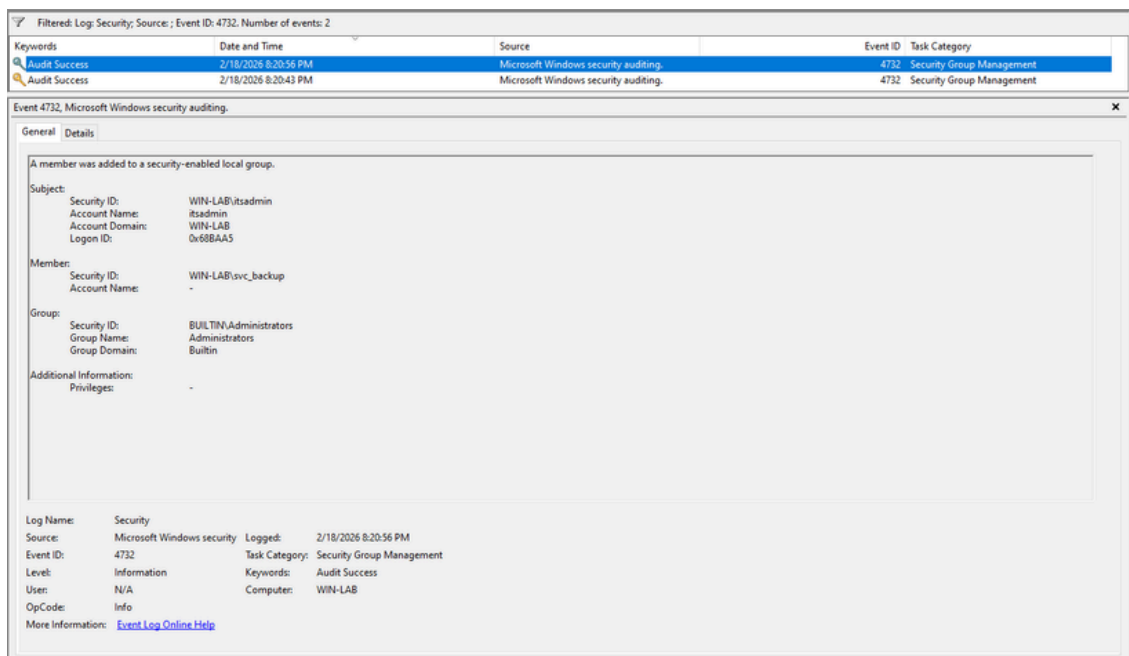
After obtaining administrative access, the attacker established persistence by creating a new local account (svc_backup) on the compromised host (WIN-LAB). The account was subsequently added to the local Administrators group, granting sustained privileged access. Event ID 4720 confirms user account creation, while Event ID 4732 indicates modification of local group membership. This sequence demonstrates deliberate persistence via privileged account establishment.



Windows Security Log – Event ID 4720
New local user account created (svc_backup).



Splunk Detection – EventCode=4720
User account creation observed on WIN-LAB.



Windows Security Log — Event ID 4732
User added to local Administrators group.



Splunk Detection — EventCode=4732
Local group membership modification detected.

Analyst Assessment:

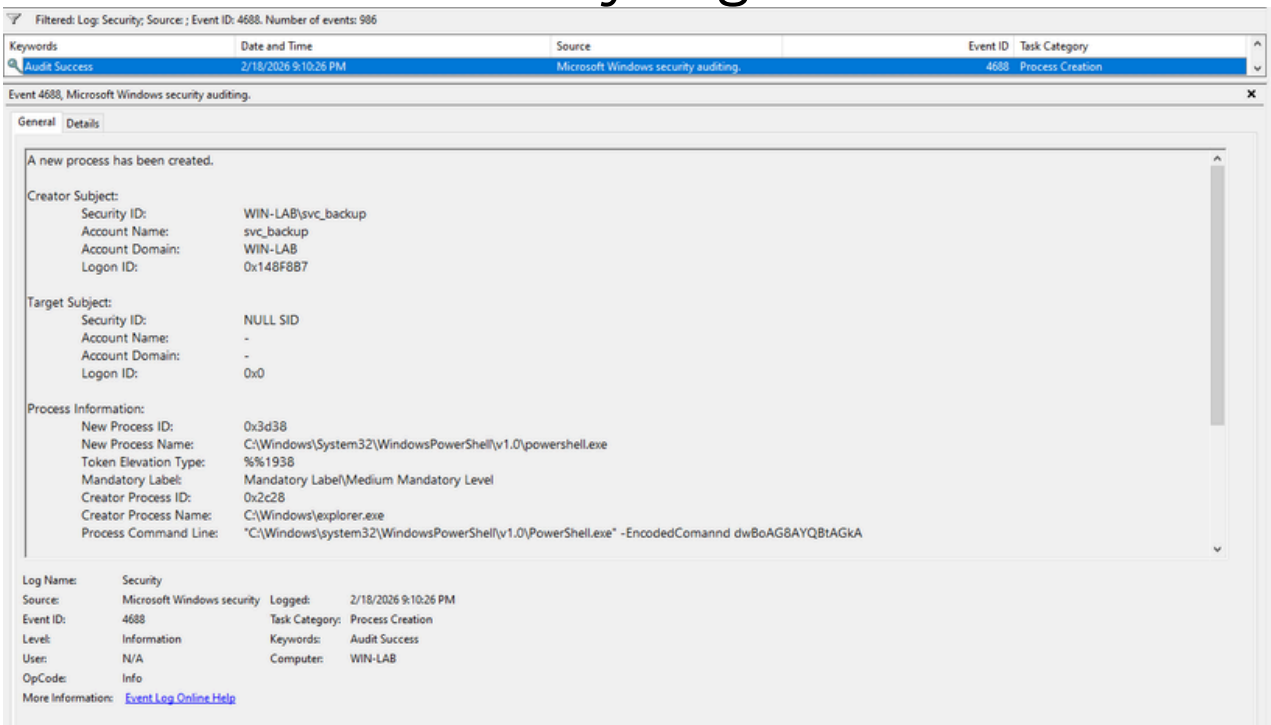
The creation of a new administrative account immediately following a privileged logon strongly indicates persistence establishment. In a production enterprise environment, this behavior would warrant immediate containment, credential reset, and audit of additional unauthorized accounts.

Phase 4 – Execution

Following persistence establishment, process creation telemetry (Windows Security Event ID 4688) captured the execution of PowerShell with encoded command parameters on the compromised host (WIN-LAB). The use of encoded arguments (-EncodedCommand) is commonly associated with post-exploitation activity and obfuscation attempts intended to bypass basic detection controls.

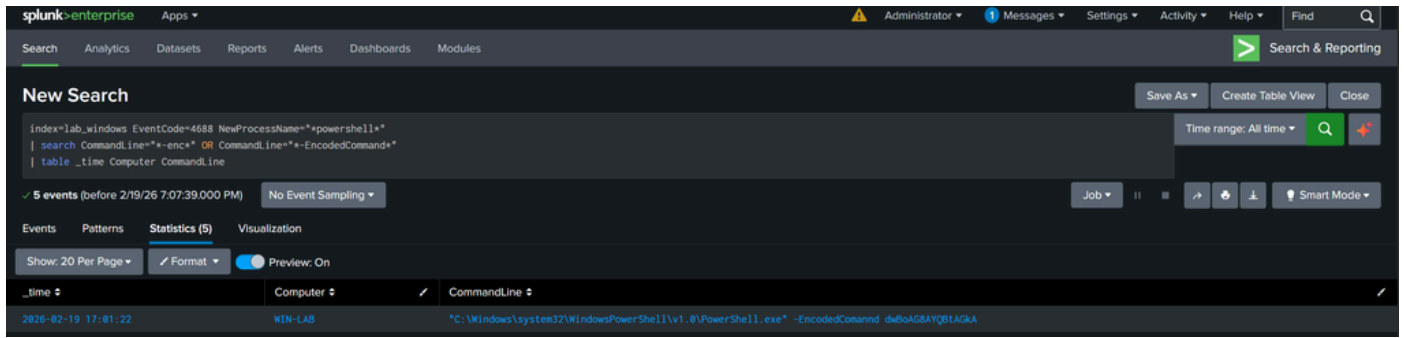
Although PowerShell Script Block Logging (Event ID 4104) was not enabled at execution time, command-line auditing provided sufficient visibility to identify suspicious execution behavior.

A. Windows Security Log – Event ID 4688



Windows Security Log – Event ID 4688
Process creation event showing PowerShell execution with encoded command argument.

B. Splunk Detection — Encoded PowerShell (4688)



Splunk Detection — EventCode=4688
Encoded PowerShell execution identified on WIN-LAB.

Analyst Assessment

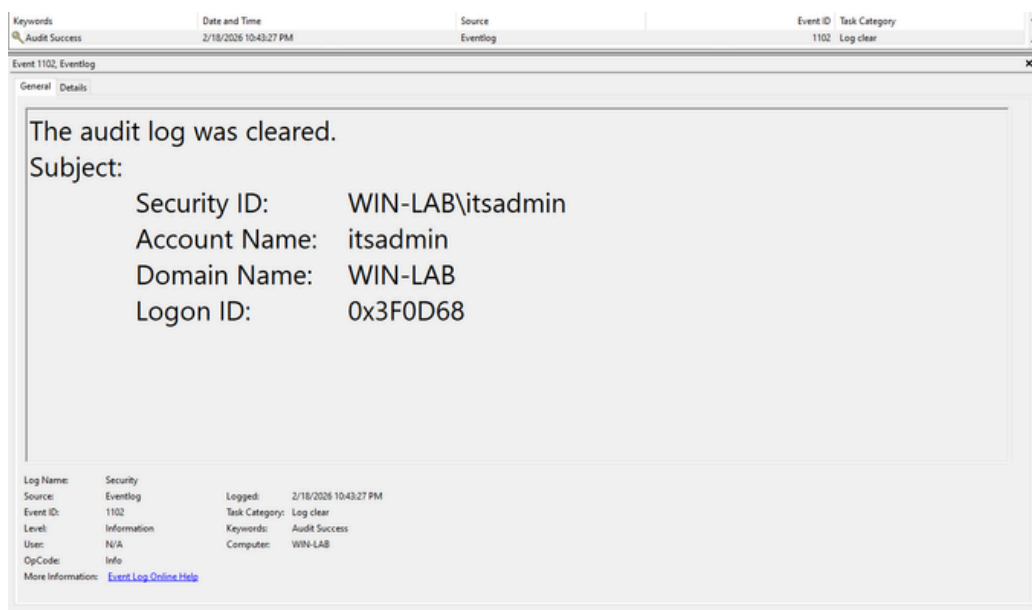
Encoded PowerShell execution is frequently leveraged by adversaries to obscure command intent and evade signature-based detection. In this case, Event ID 4688 provided sufficient visibility into the encoded command invocation. In enterprise environments, enabling PowerShell Script Block Logging (Event ID 4104) would further enhance detection capability by exposing decoded script content for analysis.

Phase 5 – Defense Evasion

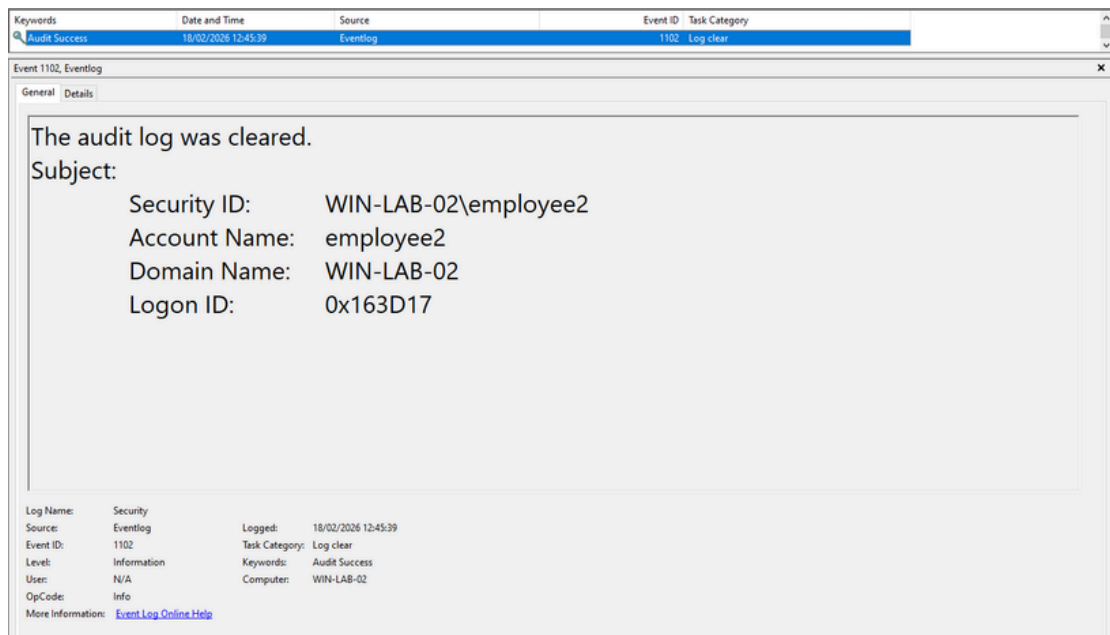
Following command execution activity, a Windows Security Log clearing event (Event ID 1102) was observed on both compromised systems. Event ID 1102 is generated when the Security log is cleared and is commonly associated with attacker attempts to remove forensic evidence and evade detection.

The timing of the log clearing shortly after privileged activity and PowerShell execution indicates deliberate post-compromise defense evasion.

A. Windows Security Log – Event ID 1102

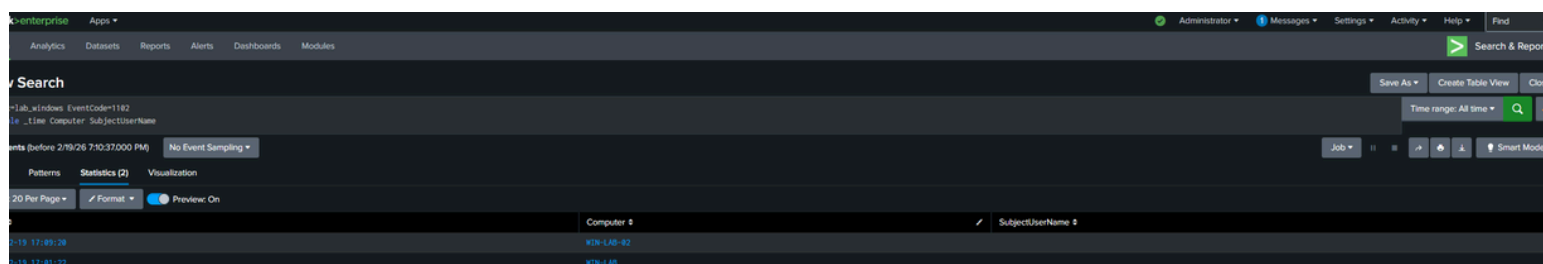


Windows Security Log – Event ID 1102
Security log cleared by privileged account. WIN-LAB-01



Windows Security Log — Event ID 1102
Security log cleared by privileged account. WIN-LAB-02

B. Splunk Detection — EventCode=1102



Splunk Detection — EventCode=1102
Security log clearing event detected on compromised hosts.

Analyst Assessment:

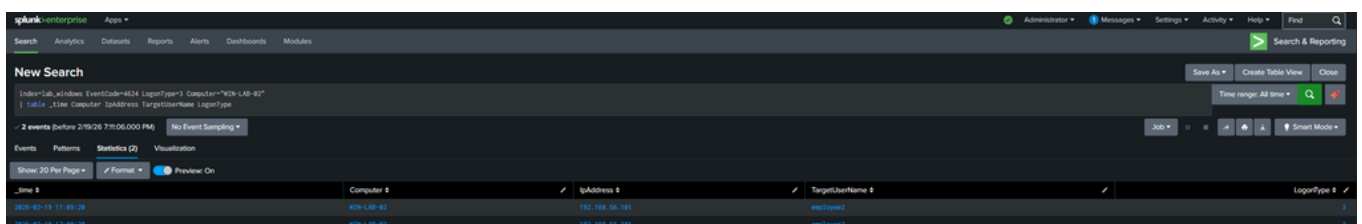
Security log clearing is a high-confidence indicator of malicious intent. Legitimate administrative operations rarely require deletion of security audit logs, particularly immediately following privileged authentication and command execution activity. In enterprise environments, Event ID 1102 should trigger immediate high-severity alerting and containment procedures.

Phase 6 – Lateral Movement

Following persistence and log tampering on WIN-LAB, a successful network logon (Event ID 4624, Logon Type 3) was observed on WIN-LAB-02 originating from 192.168.56.101 (WIN-LAB).

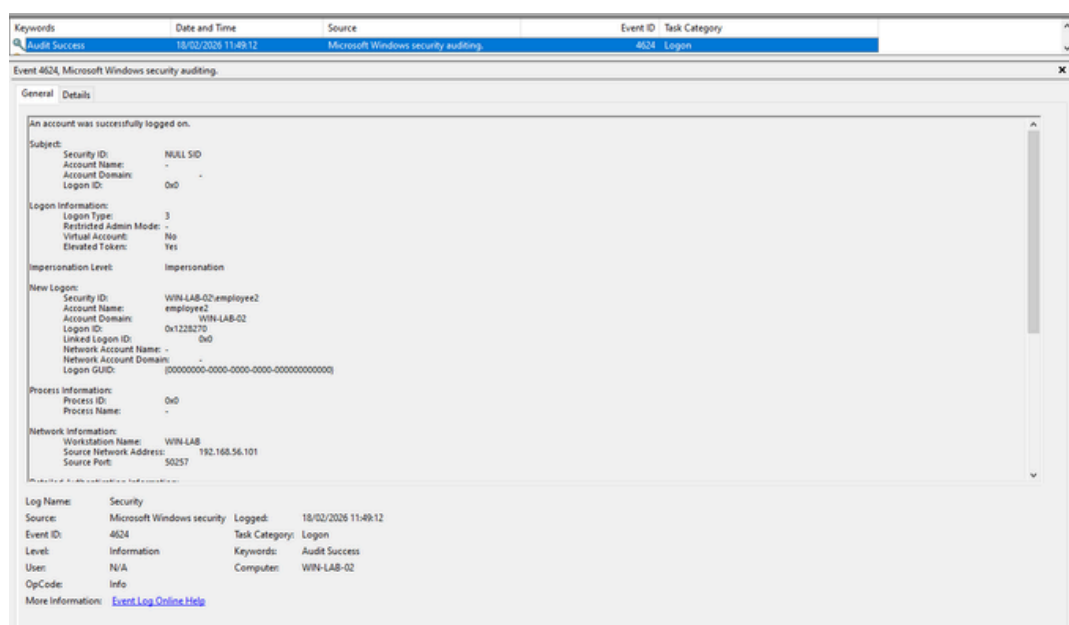
Logon Type 3 indicates a network authentication event consistent with SMB access. The activity aligns with administrative share usage (\\WIN-LAB-02\C\$) using valid credentials, confirming lateral movement via SMB.

A – Splunk Detection

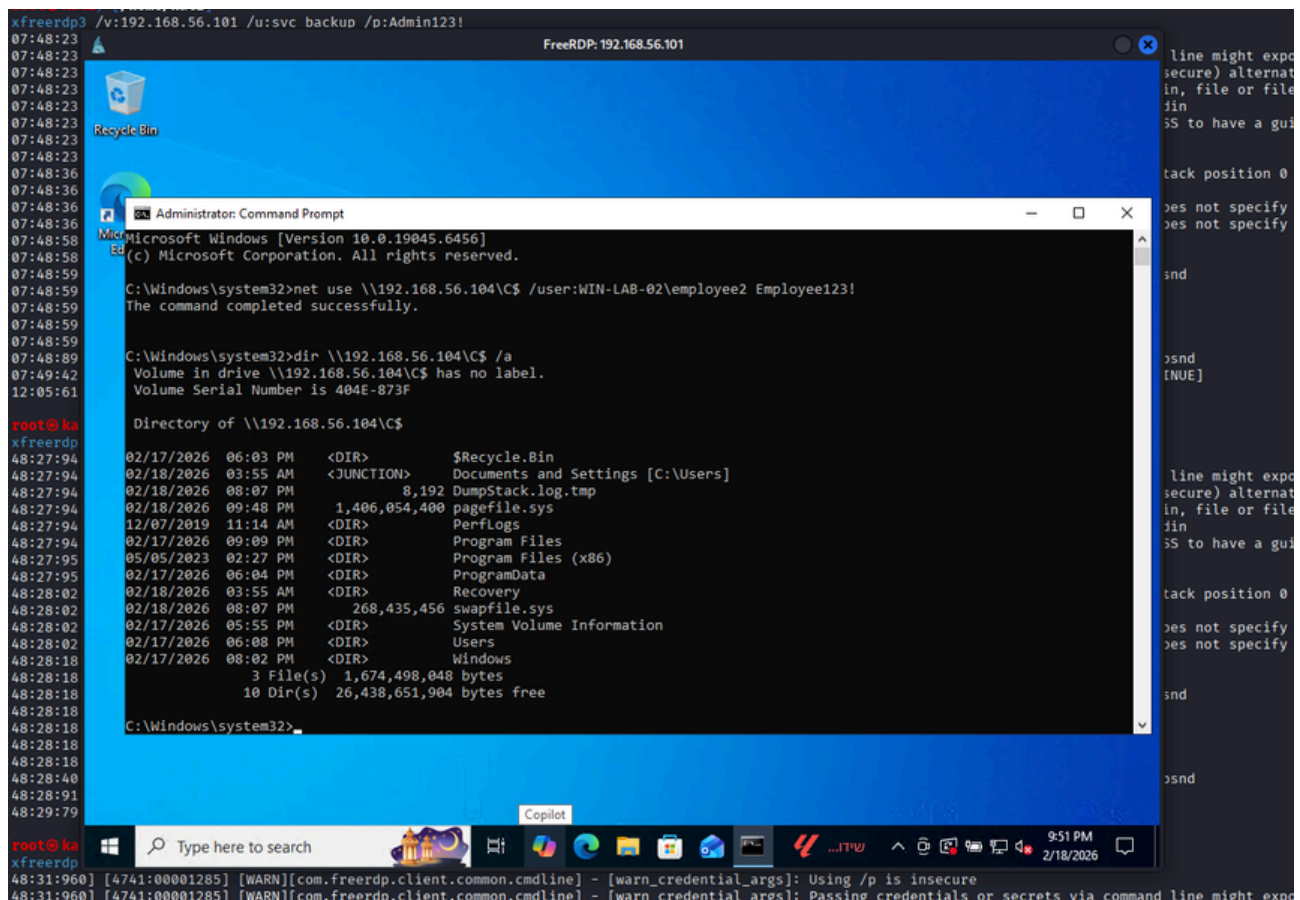


Splunk Detection – EventCode=4624 (Logon Type 3)
Successful SMB authentication on WIN-LAB-02 from compromised host WIN-LAB.

B – Windows Event Viewer Evidence



C – Attacker Command Execution Proof



Command Execution – SMB Administrative Share Access

Successful authentication and directory listing of remote C\$ administrative share.

Analyst Assessment

SMB-based lateral movement using valid credentials represents credential reuse following initial compromise. Logon Type 3 combined with administrative share access indicates post-compromise pivoting behavior consistent with ATT&CK technique T1021.002 (SMB/Windows Admin Shares).

This activity demonstrates full attack chain progression from initial access to internal host pivoting.

MITRE ATT&CK Mapping

Phase	Technique	Technique ID
Initial Access	Brute Force	T1110
Privilege Escalation	Valid Accounts	T1078
Persistence	Account Manipulation	T1098
Execution	PowerShell	T1059.001
Defense Evasion	Indicator Removal on Host	T1070.001
Lateral Movement	SMB/Windows Admin Shares	T1021.002

The observed attack chain maps to multiple MITRE ATT&CK techniques, demonstrating realistic adversary tradecraft including credential brute force, privilege escalation, persistence via account manipulation, PowerShell execution, security log tampering, and SMB-based lateral movement.

Detection Engineering Summary

- RDP brute-force detection via Event ID 4625 threshold
- Privileged logon correlation (4624 + 4672)
- Local account creation monitoring (4720)
- Administrative group modification detection (4732)
- Encoded PowerShell execution detection (4688)
- Security log clearing alert (1102)
- Lateral movement detection via 4624 Logon Type 3

All detection logic was validated in Splunk using
index=lab_windows telemetry.

Lessons Learned

Security Recommendations

- Enforce account lockout policies
- Disable local admin credential reuse
- Enable PowerShell Script Block Logging (4104)
- Monitor Event ID 1102 with high severity alerting
- Implement centralized log retention
- Apply LAPS / PAM

Incident Conclusion

This controlled simulation demonstrated a full multi-stage intrusion lifecycle, from brute-force authentication to SMB-based lateral movement. The case study validates SOC detection capabilities using Windows Security Logs and Splunk correlation. The project reflects Tier 1/Tier 2 investigative workflow and detection engineering fundamentals.